

DLR-TomoSAR — Physics-constrained learning

five loss terms · curriculum · the unrolled solver

July 2026

Why constrain the solution space

- The labels are classical fits: the network's ceiling is the label quality — and Act IV showed the network happily leans on statistical shortcuts.
- Physics terms restrict predictions to profiles **consistent with the SAR signal model**:
 - today — as a regularizer: physical quantities recomputed from the predicted profile must match those of the GT profile;
 - next — as self-supervision against the *measured* covariance itself, removing the label ceiling entirely.
- Five terms implemented and theory-tested; every one differentiable end-to-end through the Gaussian-mixture curve reconstruction.

The shared forward operator

All terms use the vertical wavenumber and the steering matrix on the elevation grid:

$$k_z^{(i)} = \frac{4\pi b_i}{\lambda r_0 s}, \quad A_{in} = e^{j k_z^{(i)} \xi_n}$$

- Geometry resolved from campaign track parameters — not nominal defaults: per-pass perpendicular baselines, look angle, slant range.
- **Per-pixel k_z field** for range/azimuth-varying geometry, built at preprocessing and threaded dataset → trainer → loss.
- Steering sign settled empirically ($+k_z$: median peak error 0.67 m, within one Rayleigh cell).

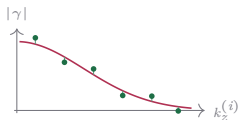
Data-consistency terms (the ranked top two)

Coherence re-synthesis

coherence domain

$$\gamma_P(k_z^{(i)}) = \frac{\sum_n P(\xi_n) e^{jk_z^{(i)} \xi_n}}{\sum_n P(\xi_n)}$$

$$\ell_{\text{coh}} = \left\langle \frac{1}{N_s} \sum_i |\gamma_P(k_z^{(i)}) - \gamma_T(k_z^{(i)})|^2 \right\rangle$$



γ_P from the predicted mixture pulled onto the target coherences,
baseline by baseline.

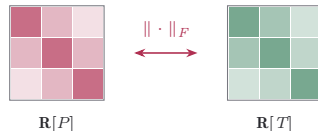
Cloude 2006 (PCT); same Fourier kernel as gamma-Net
(Qian et al.).

Covariance matching

covariance domain

$$\mathbf{R}[P] = \mathbf{A} \text{diag}(P) \mathbf{A}^H \Delta \xi$$

$$\ell_{\text{cov}} = \left\langle \frac{\|\mathbf{R}[P] - \mathbf{R}[T]\|_F^2}{\|\mathbf{R}[T]\|_F^2} \right\rangle$$



Synthesised vs target covariance — structure and relative scale,
entry by entry.

COMET (Ottersten, Stoica, Roy 1998); TomoSAR: Cros &
Ferro-Famil 2025; CS fidelity: Berenger et al. 2023.

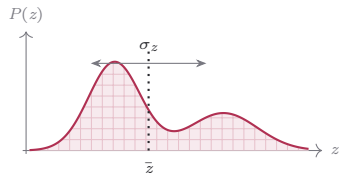
Descriptor terms: moments and total power

Moments

mass · phase-centre $\bar{z}$ · spread σ_z

$$m = \sum_n P(\xi_n) \Delta\xi$$

$$\bar{z} = \frac{\sum_n \xi_n P(\xi_n)}{\sum_n P(\xi_n)}, \quad \sigma_z^2 = \frac{\sum_n (\xi_n - \bar{z})^2 P(\xi_n)}{\sum_n P(\xi_n)}$$



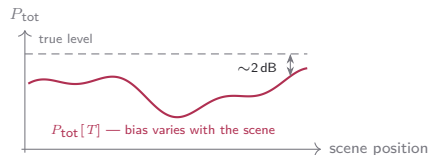
Centroid and spread are bias-robust *ratios* (Cros & Ferro-Famil 2025).

Total power

the one absolute-power quantity

$$P_{\text{tot}}[\cdot] = \sum_n (\cdot)(\xi_n) \Delta\xi$$

$$\ell_{\text{pow}} = \left\langle \left(P_{\text{tot}}[P] - P_{\text{tot}}[T] \right)^2 \right\rangle$$

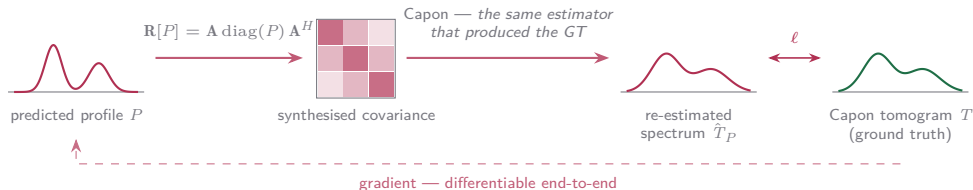


The target $P_{\text{tot}}[T]$ is not the true power: its bias moves with scene content (Lopez-Dekker & Mallorqui 2010) — the loss would imprint that bias.

The estimator-cycle term: Capon cycle

Push the prediction through the *same estimator* that produced the ground truth:

$$\hat{T}_P(\xi_n) = (\mathbf{a}^H(\xi_n)(\mathbf{R}[P] + \epsilon\bar{\sigma}\mathbf{I})^{-1}\mathbf{a}(\xi_n))^{-1}$$

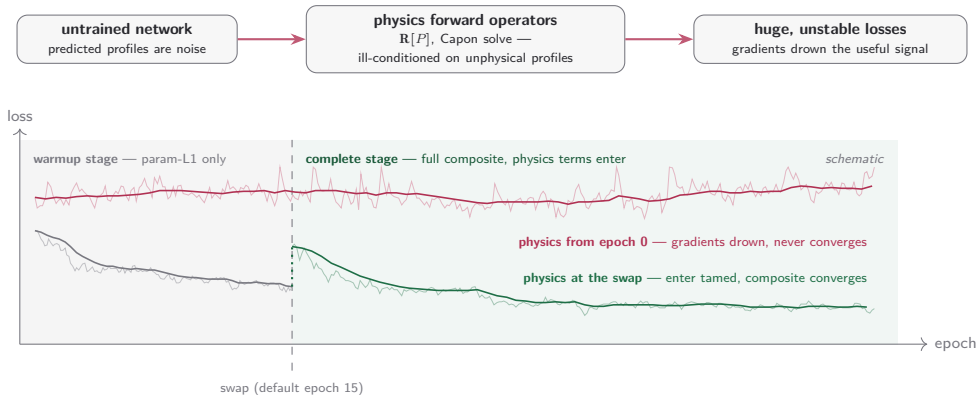


- Absorbs the Capon bias and the PSF **by construction**.
- Costs an $N_s \times N_s$ solve per pixel — the most expensive of the five.

Cycle-consistency lineage: Yaman 2020, Lu 2021, Vella & Mota 2020, Angermann 2023.

Why the physics terms need a curriculum

The physics terms are the most sensitive consumers of the reconstructed profile — and at random initialisation they see only noise:



Swap mechanics, calibration — and status

Each detail is an anticipated failure mode:

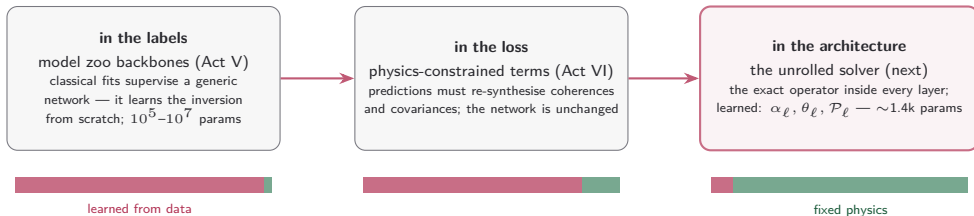
- **Resets at the swap:** LR schedule origin, LR warmup, Adam moments, early-stopping counters — optimizer state calibrated to the old loss is stale under the new one.
- **Checkpoint-baseline reset:** best-loss tracking restarts across the non-comparable loss scales — otherwise the pre-swap best could never be beaten.
- **Stage inheritance:** the complete stage is the base configuration, warmup a derived pre-phase materialised at launch — the stages cannot drift.
- **Loss-scale probe:** every active term run at unit weight over probe batches, IQR-filtered, folded into the per-term weights — physics terms enter at calibrated magnitude, not guessed weights.

Status

Implemented, SAR-theory-verified, geometry wired end-to-end — **training results pending.**

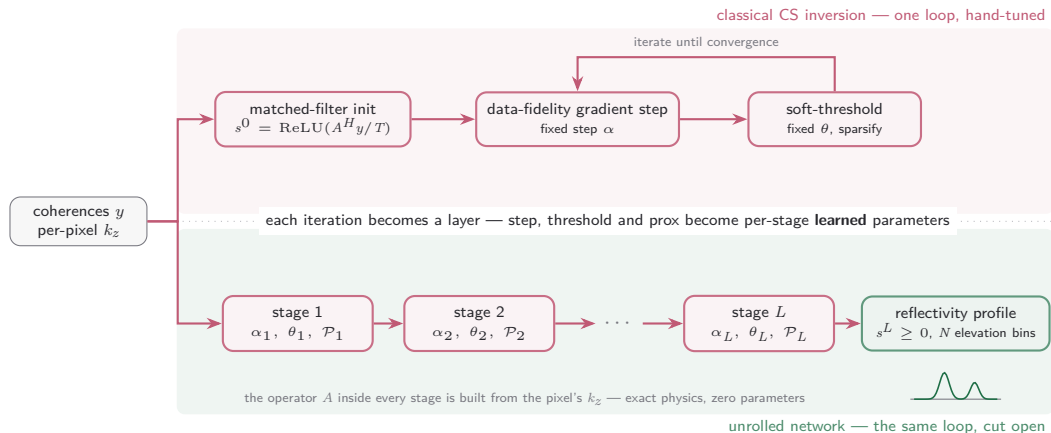
Context: where does the physics live?

Three places the SAR signal model can enter a learned inversion — the project now covers all three:



The unrolled family probes the far end of the axis — **the question it answers:** how much of the zoo's capacity is spent re-learning physics we already know?

The unrolled model: the solver becomes the network



$$r^\ell = s^\ell + \alpha_\ell \text{Re}[A^H(y - As^\ell)] / L_{\text{lip}}, \quad s^{\ell+1} = \text{ReLU}(\mathcal{P}_\ell(r^\ell) - \theta_\ell)$$

γ -Net lineage (Qian et al., IEEE TGRS 2022); LISTA (Gregor & LeCun 2010); ISTA-Net (Zhang & Ghanem 2018). $\mathcal{P}_\ell$: per-pixel residual 1D convolution along elevation; $L_{\text{lip}} = TN \text{d}z^2$ normalises the step. Default $L=8$: $\sim 1.4\text{k}$ parameters.

Adapting γ -Net — a different unknown, a per-pixel operator

γ -Net — one scene geometry, weights learned



a fixed matrix can be learned — an operator that changes at every pixel cannot



this project — per-pixel geometry, physics exact, scalars learned

γ -Net inverts $g = \mathbf{R}\gamma + \varepsilon, \quad \gamma \in \mathbb{C}^N$

this project inverts $y = A(x)s, \quad s \in \mathbb{R}_{\geq 0}^N$

g : one complex look per track (first-order); y : per-track coherences (second-order) — averaging interferograms cancels the speckle phase, so the recoverable object is the power profile s , not the complex reflectivity γ . ε : measurement noise; T : tracks; N : elevation bins; $\mathcal{P}_\ell$: per-pixel residual 1D convolution along elevation.

Adapting γ -Net — mechanism by mechanism

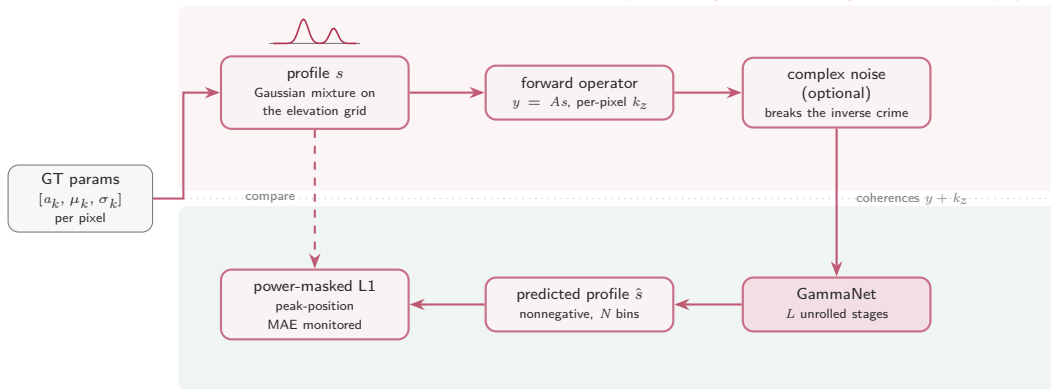
mechanism	γ -Net (Qian et al. 2022)	this project	why
iteration	gradient step $\rightarrow$ shrinkage, L stages	= kept	the unrolling premise itself
supervision	train on simulated measurements	= kept	the exact operator generates the pairs
unknown	complex reflectivity $\gamma \in \mathbb{C}^N$	$\rightarrow$ power profile $s \in \mathbb{R}_{\geq 0}^N$	speckle phase averages out of coherences
measurements	single-look stack g	$\rightarrow$ per-track coherences y	the pipeline is second-order end to end
operator	one fixed $\mathbf{R}$, scene-wide	$\rightarrow$ $A(x)$ from the pixel's k_z	airborne geometry varies across the scene
learned weights	$\mathbf{W}^\ell$ per stage, coupled to $\mathbf{R}$	$\rightarrow$ none — scalars α_ℓ, θ_ℓ	a per-pixel matrix cannot be learned
shrinkage	support selection + piecewise-linear function	$\rightarrow$ prox $\mathcal{P}_\ell$, then $\text{ReLU}(\cdot - \theta_\ell)$	the prior must respect $s \geq 0$
model order	Bayesian information criterion after inversion	$\rightarrow$ downstream parameter extraction	the pipeline already fits K scatterers

The contract every change preserves: **the physics stays exact inside the network — only the prior is learned.**

support selection: the largest-magnitude entries bypass the threshold at each stage; piecewise-linear function: γ -Net's learned replacement for soft-thresholding; coupled weights: $\mathbf{W}_2^\ell = \mathbf{I} - \mathbf{W}_1^\ell \mathbf{R}$ halves the free parameters per stage. γ -Net reaches its accuracy at 12 stages; default here $L = 8$. **green** = exact physics carried by the operator · **red** = learned · grey = kept from γ -Net.

Training the unrolled model against the exact operator

synthesis — ground truth through the exact forward physics



inversion — the unrolled network runs it backwards

Phase 1 (implemented): learn to invert the *exact* operator on synthesised coherences. Phase 2 (pending): measured covariances from the stack — no label ceiling.

Isolated family — own registry, trainer and entry point (`train_unrolled`); optimizer splits steps/thresholds from the prox; no AMP, the operator is complex-valued.

Why it exists: the physics-native comparison point — if $\sim 1.4k$ parameters of encoded physics match capacity-matched vision backbones, those backbones are mostly re-learning known physics.